

Soil carbon stocks and dynamics of different land uses in Italy using the LUCAS soil database

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ABSTRACT

In terrestrial biosphere, soil represents the largest organic carbon pool, and a small change of soil organic carbon (SOC) can significantly affect the global carbon cycle and climate. Land use change (LUC) and soil management practices coupled with climate variables can significantly influence the soil organic carbon stocks (SOC-S) and its dynamics; however, our understanding about the responses of SOC in different LUC's (e.g., cropland, grassland and forest land) to mitigate climate change is quite limited at country level like Italy. Thus, the aims of this study were which factors are affecting SOC dynamics in three LUC's over time across Italy; and their relevance in terms of SOC-S in the superficial layer of soil that significantly contributes to the climate change mitigation, using LUCAS soil database. To calculate the SOC-S, it is necessary to have soil bulk density (BD) which is not present in the LUCAS database. Thus, we estimate the soil BD using the pedotransfer function (PTFs); and results shows that the soil BD obtained from fitting of the PTFs were reasonable to estimate the SOC-S for different land use types ($R^2 \geq 0.75$). Overall, results showed that LUC's and soil management practices can significantly ($p < 0.001$) influences SOC dynamics and SOC storage from the soil and varied among LUC's but not for over time except grassland. Spatially, the mean SOC-S storage of the different LUC's was in the following order: forest land > grassland > cropland for both years 2009 and 2015. On the other hand, the SOC-S storage increased by 8.33% for cropland, 13.56% for forest land, and 29.79% for grassland during the year of 2009–2015, while SOC-S storage increased significantly ($p < 0.001$) in grassland over time but not for cropland and forest land which also follow the increasing trend but insignificantly. Our results also reveal that the SOC dynamics negatively correlated with MAT, and positively correlated with MAP for all land uses except forest land. Thus, this research indicates that LUC's and soil management practices coupled with climate variables can significantly influence SOC storage and its dynamics in the superficial layer of soil which have the potential capacity to mitigate climate change.

1. Introduction

Soil has the biggest organic carbon (OC) reservoir in the terrestrial biosphere (Batjes, 1996; Scharlemann et al., 2014; Schmidt et al., 2011), second only to oceans (Penman et al., 2003), having twice as much carbon (C) as the atmosphere and three times that stored in global vegetation (Bradford et al., 2016; Mayer et al., 2020; Viscarra Rossel et al., 2019). Globally, soil organic carbon (SOC) estimations are still being debated, and there is no precise harmony on these magnitudes, revealing a carbon content varying from 600 to 800 Pg of C stored in the top 0.2–0.3 m of soil thickness to 2300–3000 Pg of C in the top 2–3 m of soils (Guo and Gifford, 2002; Jobba'gy et al., 2000; Sanderman et al.,

2017). Also, Govers et al. (2013) estimated that mean SOC stock of 1400–1600 Pg C (1 m depth) to 1990–2460 Pg C (2 m depth), which is consistent with Batjes (1996) and Todd-Brown et al. (2013). However, even little changes in the SOC pool can drastically affect the concentration of greenhouse gases (GHGs) in the atmosphere, jeopardizing the existing climate situation (Bradford et al., 2016; Lal et al., 2018; Paus-tian et al., 2019). Simultaneously, soil carbon sequestration decreases atmospheric greenhouse gas (GHG) intensities and should have a significant role in mitigating climate change during the next century (Caddeo et al., 2019; Smith, 2004). Moreover, several meta-analyses have revealed that the foremost aspects that contribute to SOC stock (SOC-S) restoration after revegetation include: the prior land use (LU),

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the tree types planted, the soil clay content and the climate (Li et al., 2012; Olaya-Abril et al., 2017). Through the fortification of current land-based carbon stock and boosting nature-based terrestrial sequestration, policy-makers, and land managers can reduce GHGs emission and remove atmospheric CO₂, which can deliver other ecosystem services (Fryer and Williams, 2021). Utilizing Nature-Based Climate Solutions (NBCS), one-third of the cost effective GHGs mitigation needed by 2030 to keep warming below 2 °C may be achieved (Griscom et al., 2017). Though the offset capacity to lower the global C budget remains uncertain, the greatest potential of NBCS is anticipated to be 23.8 Pg CO₂ yr⁻¹ (6.49 Gt yr⁻¹) by 2030, with soil carbon accounting for 25% of this potential (Bossio et al., 2020).

Soils have historically lost from 40 to 90 Pg C altogether due to farming and disturbance (Houghton et al., 1999; Lal, 2004). Also, it is projected that during the 1990s, land use change (LUC) released 1.6 ± 0.8 Pg C y⁻¹ into the atmosphere (IPCC et al., 2001; Schimel et al., 2001). However, LUC significantly affects SOC stocks (Guo and Gifford, 2002), and most long-term LUC experiments demonstrate a similar effect on SOC (Smith, 2004; Smith et al., 2002). This is projected to continue; a recent modelling research investigating the possible effects of climate and LUC on SOC stocks in Europe revealed that LUC has a greater net influence on SOC storage than anticipated climate change (Smith et al., 2005). A long-term experiment of meta-analysis revealed that transforming forestry or grassland to croplands resulted in considerable SOC loss, but converting forest land to grassland did not result in SOC loss under all circumstances (Guo and Gifford, 2002). Moreover, the overall ecosystem C incorporating above-ground biomass decreases as a result of the loss of tree biomass carbon. In Brazil, parallel findings were documented, where overall ecosystem C losses are enormous but soil C remains stable (De Moraes et al., 1996; Lal et al., 2018), despite the fact that previous research have demonstrated that SOC depletion during the transition from forest to grassland (Detwiler and Hall, 1988; Olson, 2007). Even in the best-case scenario, only around 10 percent of the overall ecosystem C misplaced due to anthropogenic activities like deforestation (e.g., tree removal, fire, etc), can be regained (Smith et al., 2019). In brief, SOC is typically lost when grasslands, woodlands, or other natural ecosystems are converted to croplands, or when highly organic soils are drained, cultivated, or limed. By contrast, SOC contents tend to rise when former croplands are converted to grasslands, woodlands, or native vegetation, or when organic soils are returned to their natural state. Best management techniques that boost C inputs to the soil (e.g., improved manure and residue management) or minimize losses (e.g., lower impact tillage, reduced residue removal) assist to preserve or upsurge SOC quantities, regardless of where the land is managed.

Forest soils alter the microclimate conditions, particularly the temperature and moisture regimes, and enhance organic matter (OM) inputs, so enhancing the soil's quality (Bouwman and Leemans, 1995). Also, the no-tillage state in this ecosystem boosts soil microbial and fungal populations, as well as the stable aggregate formation (Jégou et al., 2000), which protects against OM breakdown (Del Galdo et al., 2003). Therefore, Mediterranean forest soils have a high capacity for C sequestration and storage (Roig and Rubio, 2009). So, the forest soils are performing a significant role in the global C balance (Bayer et al., 2017). But we must not overlook the fact that soil C storage is reducing on a worldwide basis (Jandl et al., 2007) and that Mediterranean forest are susceptible to C storage forfeiture (Badalamenti et al., 2017). The proportion of C stored in forest soils accounts for a significant fraction of total C storage worldwide. Six et al. (2002) predicted that SOC stored in forest soils accounted for 70–73 percent of total world SOC quantities. However recent times, Pan et al. (2011) estimated worldwide forest C sinks and projected total SOC stock in 861 Pg (43 Pg: 5% in litter; 73 Pg: 8% in deadwood; 363 Pg: 42% in below and above-ground biomass; and 383 Pg: 45% in soil at 1 m depth). By contrast, Chiti et al. (2018) found that woody encroachment on former croplands and pastures mostly helps to climate change mitigation, by raising the initial SOC stock of 120% (66.5 Mg C ha⁻¹) and 45% (40.5 Mg C ha⁻¹), respectively.

Moreover, the LUC from cropland to grassland significantly contributes to the mitigation of climate change by increasing SOC of around 80% of the initial SOC (45.9 Mg C ha⁻¹).

Many factors influence SOC dynamics, including environmental constraints, climate variability, vegetation growing, and sometimes even anthropogenic activities (Gaitán et al., 2019; Ramesh et al., 2019), therefore it is difficult to get precise spatial and temporal aspects of SOC storage at a wide scale throughout time (Dignac et al., 2017; Liu et al., 2011, 2019). Therefore, there is indeed a clear connection between “time” and “soil formation processes” (Stevens and Walker, 1970), emphasizing the relationship between time, land management, and LUC due to man-made influences in terms of soil evolution (Montanarella et al., 2016). In 2015, the land use, land use change and forestry sector (LULUCF) in Italy was responsible for a net sink of 36,218 Gg CO₂ eq., which rose on average by roughly 2,000 Gg CO₂ eq. each year, owing primarily to the expansion of grassland and forest area (NIR, 2017). In precise, between 1996 and 2015, the alteration of 1.3 million ha from grassland to forest land resulted in a sink of 1937 Gg C ya⁻¹, of which 242 Gg C ya⁻¹ is allocated to the SOC pools. Furthermore, due to the peculiar geological and geomorphological conditions in Italy, SOC variability is quite high, and soil surveying with such variability is difficult and onerous (Marchetti et al., 2012). SOC degradation is stronger in locations with Mediterranean climates, such as central and southern Italy, due to the combined impact of high temperatures and low soil moisture, which increases the mineralization rate.

To our knowledge, LUCAS (Land Use and Coverage Area frame Survey) topsoil is indeed one of the largest global and most comprehensive, integrated continental-scale soil inventories due to the wide range of properties investigated (Orgiazzi et al., 2018). Likewise, it is also a fully accessible repository tool that contains data from the European Soil Data Centre (ESDAC) that is freely available (Panagos et al., 2012). This dataset provides a unique source of information for researchers; not only geochemical and pedological analyses but also ecological and biological studies (Crowther et al., 2016), which can benefit from the LUCAS Soil database (Ettema and Wardle, 2002). LUCAS Soil-derived outcomes are also significant tools that policy-makers utilize to support decision-making processes connected to the agricultural and environmental concerns (Keesstra et al., 2016). So, this dataset (LUCAS) could be a good option for evaluating the status of SOC and determine SOC dynamics overtime at the country level. However, owing to a lack of reliable data at the country level, the SOC and SOC stock (SOC-S) assessments in the National GHG Inventory are rather sparse, using a quantitative measure of SOC content for the entire Italy in the forest land, grassland, and cropland categories, respectively. For this reason, the main goal of this study is to quantify over time the effects in the SOC stocks, evaluating three LUC's to assess the LUC impacts on SOC dynamics across Italy, and its involvement in the terrestrial carbon (C) storage. And the precise aims of this research were: (a) which factors are affecting SOC dynamics in three LUC's (cropland, grassland, and forest land) over time across Italy; and (b) their consequence in terms of SOC stock (SOC-S), which substantially contributes to climate change mitigation.

2. Materials and methods

2.1. Study area and bioclimatic classification

The research region encompasses all of Italy's territory, covering latitudes of 35°47' to 46°60' N and longitudes of 5°92' to 21°39' E, while the Italian land area is roughly 300,192 km². The Italian peninsula exhibits significant climate variation and variability within a kilometre (km), due to its topographical and orographic complexity. In addition, Italy is located in the heart of the Mediterranean basin, which is combined with an extensive latitudinal range that represents a long narrow bridge between temperate and Mediterranean macrobioclimates (Biondi and Baldoni, 1994; Pesaresi et al., 2014, 2017). Indeed, Italy has both

temperate and Mediterranean macrobioclimates (Fig. 1). However, the features of the Italian area, such as geographical location, mean annual temperature (MAT), and mean annual precipitation (MAP), are depicted in Fig. 1. These data were derived from WorldClim dataset (climate surface for global land areas) with 1 km spatial resolution (Fick and Hijmans, 2017). Also, the existence of complex orographic systems, such as the Alps in northern Italy and the Apennines chain that covers the length of the Italian peninsula, as well as the vicinity of the African and Eurasian continental masses, contribute to the high level of climate variety (Kottek et al., 2006). The climate setting is crucial in the degradation processes of organic residue and the stability of organic substances (Lützow et al., 2006). In particular, the regional climatic context enhances the decomposition and mineralization of organic matter (OM).

2.2. LUCAS topsoil database

This study is based on the pan-European Land Use/Land Cover Area Frame Survey (LUCAS) topsoil database, which is managed by EUROSTAT in collaboration with the European Commission's Directorate-General for Environment and the Joint Research Centre in Ispra, Italy

(Orgiazzi et al., 2018; Tóth et al., 2013). LUCAS is the first attempt to build a comprehensive soil databank at the continental level to assist policymakers. The LUCAS survey's core component is land use (LU), hence the locations for soil sampling were based on this component. Likewise, one of the major goals of the LUCAS strategy is to develop a completely standardized database on land use/cover across the European Union and to monitor physicochemical status evolves over time. According to CORINE Land Cover classification (CLC, 2006) dataset, the number of sampling points selected was proportionate to both the surface area of each country and the percentage of each type of land use and cover in each country. The density of LUCAS topsoil (0–20 cm) sample sites, on the other hand, is about one every 199 km², allowing for a grid cell size of around 14 km (Ballabio et al., 2016).

A total of 19,967 soil samples were obtained in the framework of the LUCAS 2009 Topsoil survey for 25 EU Member States (EU-27 excluding Bulgaria and Romania) with precise geographical coordinates (Tóth et al., 2013). Also, the second LUCAS 2015 Topsoil (0–20 cm) survey is now available for research purposes where the total number of selected locations was approximately 26,000 for 28 EU countries (Orgiazzi et al., 2018). The third LUCAS soil survey, which was planned for 2018 is underway, with data expected to be published by the end of 2021.

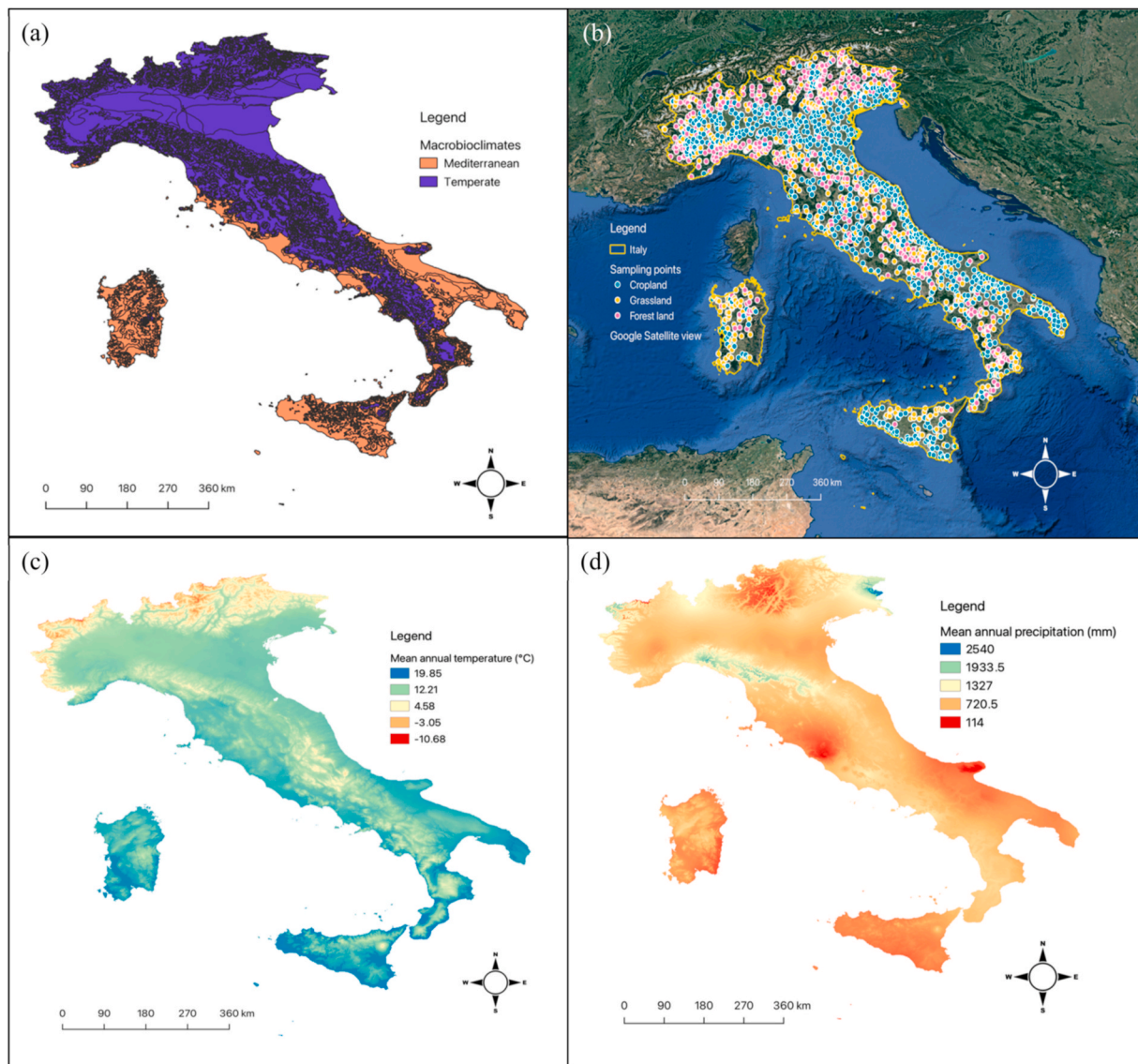


Fig. 1. (a) Macrobioclimates of the Italian territory (Temperate and Mediterranean); (b) Geographical location of the sampling points in the study area; (c) Map of the mean annual temperature (°C) over study area; and (d) Map of the mean annual precipitation (mm) in the study area.

However, in this study, we used data from a survey that took place in Italy in 2009 and 2015, with 1,278 and 1,557 top-soil (0–20 cm) samples, respectively which were categorized based on different land use types (Tóth et al., 2013). Each soil sample is classified based on its geographical coordinates (Fig. 1), as well as a detailed description of the vegetation, soil type, elevation, and physical and chemical attributes of soil (e.g., Soil C content, pH, Stone content and soil texture).

2.3. Soil organic carbon stock calculation

The amount of the total SOC stock (SOC-S) was calculated for each sample of the study area under different land cover (e.g., forest land, grassland, and cropland) using Eq. (1) (Boone et al., 1999):

$$y = a \cdot b \cdot c \cdot d \quad (1)$$

where y is the SOC stock per unit area (t C h^{-1} soil), a is the C concentration in the soil sample (g C kg^{-1} soil), b is the soil bulk density (Mg soil m^{-3}), c is the horizon depth (cm), and d is the percentage of rock fragments in the mass [1- (% rock volume/100)]. The SOC content is estimated in accordance with the ISO 10694:1995 procedure (ISO, 1995). Concisely, the total C content of a soil sample is assessed using an elemental analyser after dry combustion. After adjusting for carbonate content in the sample, the organic C content is estimated from this content. While the total C content equals the organic C content of the sample in the absence of carbonates.

2.4. Soil bulk density calculation

To compute the SOC stocks (SOC-S), it is necessary to have soil bulk density which is not present in the LUCAS database. Thus, we need to estimate the soil bulk density using the pedotransfer function (PTFs) suggested by Adams (1973) for forest land and Chiti et al. (2018a) for grassland (pasture and shrubland) and cropland (annual crops, olive

Table 1

List of pedotransfer function (PTFs) equations (linear and non-linear models) and the number of soil samples (n).

Land use	n	Pedotransfer function (PTFs)
Equation for bulk density estimation based on soil organic carbon and mineral fractions		
Annual crop	609	$BD = 1.317 - (0.054 \times \ln \text{ Clay\%}) + (0.055 \times \ln \text{ Sand\%}) - (0.054 \times \ln \text{ SOC\%})$
Olive trees	127	$BD = 1.169 - (0.047 \times \ln \text{ Clay\%}) - (0.053 \times \ln \text{ Sand\%}) + (0.100 \times \ln \text{ SOC\%})$
Vineyard	103	$BD = 1.342 - (0.040 \times \ln \text{ Clay\%}) + (0.038 \times \ln \text{ Sand\%}) + (0.011 \times \ln \text{ SOC\%})$
Fruit trees	140	$BD = 1.158 - (0.086 \times \ln \text{ Clay\%}) + (0.092 \times \ln \text{ Sand\%}) - (0.067 \times \ln \text{ SOC\%})$
Pasture	410	$BD = 1.564 - (0.056 \times \ln \text{ Clay\%}) - (0.026 \times \ln \text{ Sand\%}) - (0.097 \times \ln \text{ SOC\%})$
Shrubland	77	$BD = 1.350 - (0.054 \times \ln \text{ Clay\%}) + (0.023 \times \ln \text{ Sand\%}) - (0.094 \times \ln \text{ SOC\%})$
Equation for bulk density estimation based on soil organic carbon		
Annual crop	404	$BD = (0.049 \times \% \text{SOC} + 0.687)^{-1}$
Olive trees	83	$BD = -0.115 \times \ln(\text{SOC\%}) + 1.389$
Vineyard	70	$BD = 0.010 \times \ln(\text{SOC\%}) + 1.369$
Fruit trees	11	$BD = -0.048 \times \ln(\text{SOC\%}) + 1.281$
Pasture	213	$BD = (0.025 \times \text{SOC\%}) + 0.749)^{-1}$
Shrubland	23	$BD = -0.102 \times \ln(\text{SOC\%}) + 1.278$
Forest	540	$BD = (0.720 + 0.081 \times \% \text{SOC})^{-1}$

Note: BD is the soil bulk density (Mg soil m^{-3}), Clay% is the percentage of clay in the specific depth, Sand% is the percentage of sand in the specific depth, and SOC% is the soil C concentration in the specific depth. Here, the percentage of clay and sand are the mineral fractions.

trees, vineyard, and fruit trees), as reported in Table 1. The different PTFs approaches were followed depending on the data availability in the LUCAS database. PTFs are empirical predictive relationships based on easily accessible soil properties, namely mineral fraction (sand, silt, clay) and soil organic carbon (SOC) concentration, evaluated on disturbed soil samples (Bondi et al., 2018; Nasta et al., 2020; Patton et al., 2019). PTFs, on the other hand, are acquire using statistical methods with site-specific correlations, resulting in very erroneous predictions when employed outside of the environmental setting in which they were developed (Nasta et al., 2013).

However, the amount of sand, silt, and clay in mineral soil material was determined using laser diffraction in accordance with ISO (International Organization for Standardization) protocol number 13320:2009 (ISO, 2009). A laser diffraction analyser can detect mineral fractions (particle sizes) ranging from $0.02 \mu\text{m}$ to 2.0 mm . The LUCAS database's particle size distribution (PSD) was evaluated using two or three subsamples. Only when the first two samples were significantly different was the third sample taken.

2.4.1. Cross-validation of predicted pedotransfer functions

To calibrate the pedotransfer function (PTFs) for forest land, grassland (pasture and shrubland), and cropland (annual crops, olive trees, vineyard, and fruit trees), a dataset of 10,132, 415, and 91 samples having the data of mineral fraction (sand, silt, clay), the SOC concentrations, and the measured bulk density was taken from different sources for the 0–20 cm depths of mineral soil. This bulk density dataset comprises profiles from many key soil databases, including MediNet (Chiti et al., 2018), WoSIS (Batjes et al., 2017), and ISRIC (Tempel et al., 1996), which store soil profile data at European and global scale, respectively. The consistent over- or under-prediction of bulk density calculated with different PTFs was assessed by comparing observed and predicted values (Moffat et al., 2007) and was taken into account when evaluating the inaccuracy on SOC calculations. As a result, we chose four statistical performance indicators to assess the predictive capability of the tested PTFs: the mean absolute error (MAE), the mean difference error (MD), the Root Mean Square Error (RMSE) of prediction, which combines bias and lack of precision, and the coefficient of determination (R^2), which measures how well the data pairs fit to a (1:1) line (De Vos et al., 2005). The following are the definitions of the aforementioned indicators:

$$MAE = \frac{1}{n} \sum_{i=1}^n |P_i - O_i| \quad (2)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (O_i - P_i)^2} \quad (3)$$

$$R^2 = \frac{\sum_{i=1}^n (O_i - \bar{O})(P_i - \bar{P})}{\sum_{i=1}^n (O_i - \bar{O})^2} \quad (4)$$

$$MD = \frac{1}{n} \sum_{i=1}^n (P_i - O_i) \quad (5)$$

Where O , $\bar{O}$ and P are the observed, mean of observed, and predicted soil bulk density (BD) values, respectively. The soil sample counter is denoted by the subscript i , while n is the total number of soil samples. For the best prediction, RMSE should be as low as feasible and R^2 should be as near to one as possible (De Vos et al., 2005).

2.5. Statistical analysis

All statistical analyses and illustrations were developed using R 3.6.1 for Windows (R Core Team, 2020), first testing their normal distribution (Kolmogorov test, $p < 0.05$), and homoscedasticity and then completing the analysis of variance (two-way ANOVA), Tukey's test, and

correlations. Variations in SOC contents and SOC stocks (SOC-S) were determined among all the land use types (forest land, grassland, and cropland) and within forest land, grassland and cropland using the analysis of variance (ANOVA) test with the help of 'car' package (Fox and Weisberg, 2019). A multiple comparison test (Tukey HSD) was performed when significant differences were observed. The statistical significance threshold for the ANOVA and Tukey's test was adjusted at $p < 0.05$. Also, stepwise multiple linear regression analysis was performed to evaluate the relationship between SOC stocks (SOC-S) and soil physical (clay, silt, sand, coarse and rock fragments) and chemical (bulk density, pH, Total N, P, CaCO_3 and elevation) parameters and years. SOC stocks (SOC-S) were regarded as the dependent variable, whereas all other factors were independent variables. To minimize the number of explanatory variables and the dimensionality of the dataset, correlation analysis and principal component analysis (PCA) were undertaken. Because the PCA was based on a correlation matrix, the explanatory variables were scaled (normalized, Kolmogorov test) using their mean and standard deviation, which was then used to estimate the principal component (PC) and loadings scores (Legendre and Legendre, 2012). The PCs are formed in orthogonal directions where they discover the highest variation in the data, thus PC1 explains more variance than PC2, and so on. The proportion of total variance explained by a PC is used to represent the amount of variation explained by it. The graphic output of the SOC-S assessment was generated using the 'ggplot2' package.

3. Results

3.1. Statistical evaluation of predicted pedotransfer functions

The fits of the used (model) pedotransfer functions (PTFs) for predicting soil bulk density (Mg soil m^{-3}) were based on the values of the cross-validation coefficients of determination near the coefficients of the straight line 1:1 (Fig. 2), the lower values of mean absolute error (MAE), and the higher values of coefficient of determination (R^2) (Fig. 2), according to the statistics (De Vos et al., 2005; Nasta et al., 2020). The pedotransfer functions (PTFs) have been fitted to the variable as soil bulk density (BD), suggesting that the estimation error of soil BD is less erratic on a small or large scale. This also shows that the soil bulk density obtained from fitting of the PTFs values were reasonable to estimate the soil organic carbon stock (SOC-S) for diverse land use types (LUT). Thus, the fitted PTFs could be alternative methods to fill in data gaps for volumetric evaluations of soil BD for calculating SOC-S.

The model performance indices (MAE, RMSE, R^2 and MD) used to

assess the quality of soil BD prediction are presented in Fig. 2 for the 0–20 cm depths of soil. Also, the goodness between the predicted and measured BD data is shown in Fig. 2. However, the coefficient of determination (R^2) values are slightly high overall ($R^2 \geq 0.75$), with values of 0.79 for forest land, 0.82 for grassland, and 0.87 for cropland, respectively. Also, the comparison of the other statistical outputs chosen for PTFs evaluation shows good agreement with the measurement dataset while the MAE and the RMSE values are weak for all types of land use. The error in estimating the soil BD, as determined by calculating the mean difference error (MD) between measured and predicted values (Fig. 2), indicates a small possible over or underestimation of the real bulk density values for cropland ($0.018 \text{ Mg soil m}^{-3}$), grassland ($0.036 \text{ Mg soil m}^{-3}$), and forest land ($-0.111 \text{ Mg soil m}^{-3}$), respectively.

3.2. Relationship of soil organic carbon with bulk density

Soil organic carbon stocks (SOC-S) are not only affected by SOC contents, but also by soil bulk density (BD), indicating a significant link between SOC-S and BD. In this study, the relationship between SOC and BD was estimated, and SOC and BD showed a strong correlation. There was a statistically significant relationship between SOC and BD ($p < 0.001$), but which is negative for all land use, also for different years (Fig. 3). Although, the correlation between SOC and BD was inversely ($p < 0.001$) related for all land use systems, but became the more significant for forest land ($R = -0.94$, $p < 0.001$ for 2009 and $R = -0.86$, $p < 0.001$ for 2015) in comparison to grassland ($R = -0.79$, $p < 0.001$ for 2009 and $R = -0.78$, $p < 0.001$ for 2015) and cropland ($R = -0.13$, $p < 0.001$ for 2009 and $R = -0.48$, $p < 0.001$ for 2015). Therefore, it is true that an increase in BD will be caused a SOC content decrease. Also, this finding implies that an increase in BD is directly related to a reduction in soil ability to acquire and/or retain fresh inputs of SOC.

3.3. Effects of temperature and precipitation on the SOC in different land use systems

Soil organic carbon (SOC) and climate variables are highly interdependent (e.g., temperature and precipitation). Our findings show that SOC content was positively correlated with mean annual precipitation (MAP) and negatively correlated with mean annual temperature (MAT) for all land uses except forest land (Fig. 4). The most important climatic effect was that higher air temperatures linked to lower SOC levels, possibly due to SOC mineralization. But land use and management have a greater influence on SOC than climate (Figure S1). Although, the

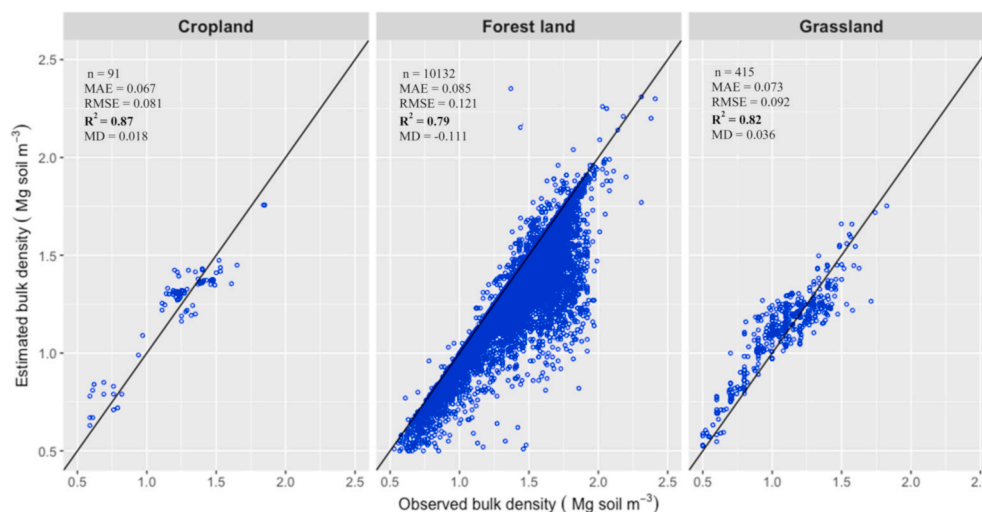


Fig. 2. The goodness of fit of pedotransfer functions (PTFs) for predicted soil bulk density (Mg soil m^{-3}) using the cross-validation data set. The straight lines represent 1:1 proportion, i.e. difference between observed and predicted (estimated) soil bulk density (Mg soil m^{-3}) using the pedotransfer functions (PTFS).

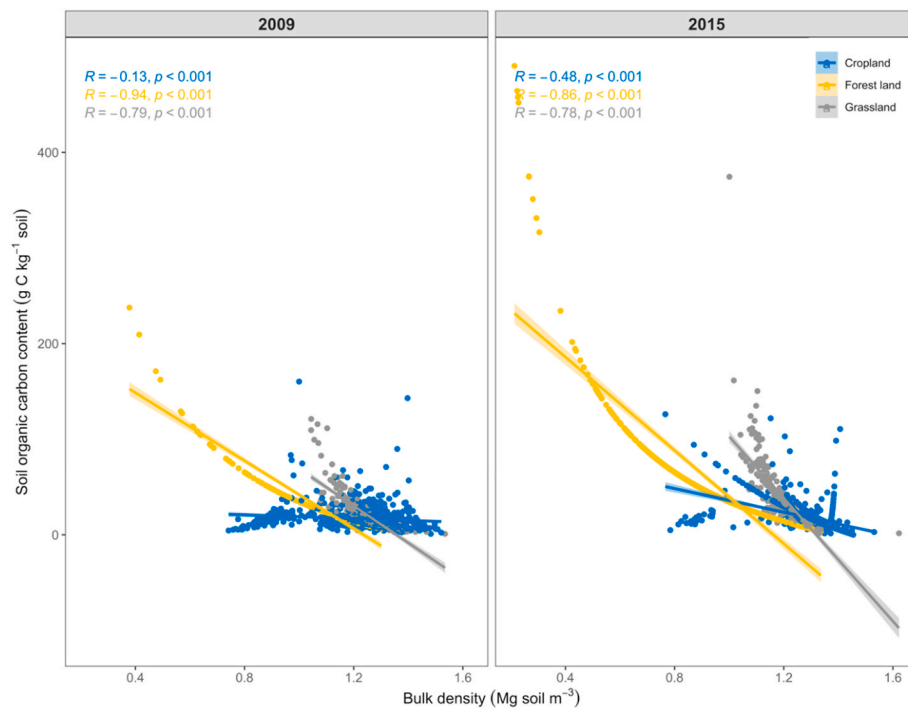


Fig. 3. Relationship between bulk density (Mg soil m^{-3}) and SOC content (g C kg^{-1} soil) for the 0–20 cm depth of soils. Circles are the experimental values from different locations of sampling sites.

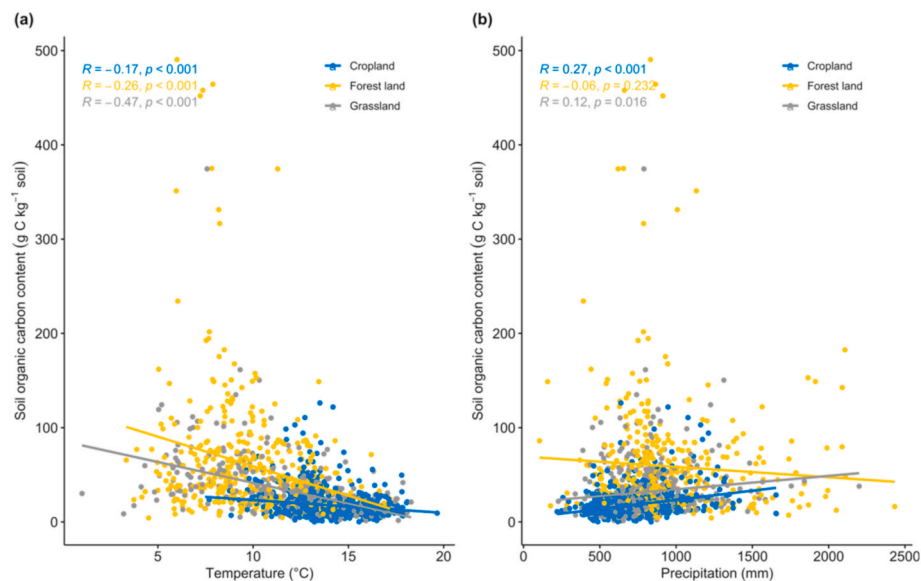


Fig. 4. (a) Relationship between the SOC content (g C kg^{-1} soil) and mean annual temperature ($^{\circ}\text{C}$). (b) Relationship between the SOC content (g C kg^{-1} soil) and mean annual precipitation (mm), for the 0–20 cm depth of soil for different land use types (Cropland, Forest land, and Grassland).

correlation with MAP was direct for cropland ($R = 0.27, p < 0.001$) and grassland ($R = 0.12, p = 0.016$), but reversed with forest land ($R = -0.06, p = 0.023$) uses (Fig. 4). As a result, the most significant climatic effect was an increase in MAT, which resulted in a decrease in SOC content. Furthermore, the influence of climate (MAT and MAP) factors on SOC content might be positive or negative depending on land use (Figure S1 and S2). Indeed, the low SOC content of lands in hot and humid climates is well recognized, and in this case of Italy, this may be a soil-based source of evidence for the magnitudes of climate tropicalization.

3.4. Effects of elevation on the SOC–S storage in different land use systems

The SOC–S storage was positively correlated with elevation for different land use systems considered in this study (Fig. 5). Statistically significant ($p < 0.001$) linear regressions were found for forest land and grassland, while for cropland was insignificant. However, the highest significant ($p < 0.001$) positive correlation was observed in grassland ($R = 0.43, p < 0.001$), followed by forest land ($R = 0.29, p < 0.001$), while cropland showed an insignificant ($R = 0.001, p = 0.752$) and negligible positive correlation (Fig. 5). Thus, our results confirmed that elevation is

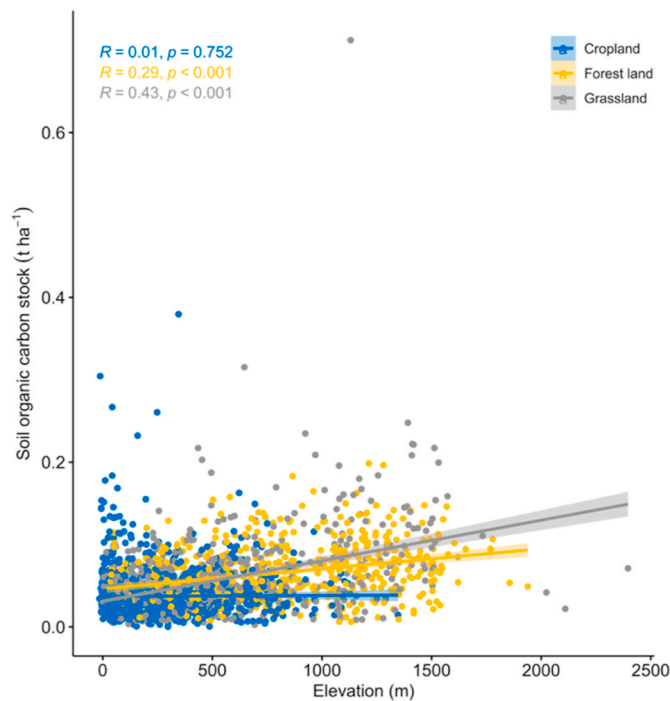


Fig. 5. The relationship between elevation (m) and SOC-S (t ha^{-1}) of different land use types.

a significant and simple predictor of SOC-S storage.

3.5. Soil organic carbon stock within cropland types and over time

All of the traits among the different cropland types, the SOC-S differed across the cropland types and over time which was significant ($p < 0.001$) within cropland types and insignificant over time (Table S1 and Fig. 6). In 2009, the storage of SOC-S was higher in perennial crops ($0.037 \pm 0.002 \text{ t ha}^{-1}$), compared to annual crops ($0.036 \pm 0.001 \text{ t ha}^{-1}$),

which were insignificant. By contrast, in 2015, the storage of SOC-S was significantly ($p < 0.001$) different within the cropland category, where perennial crops ($0.042 \pm 0.002 \text{ t ha}^{-1}$) stored the largest amount of SOC and annual crops ($0.037 \pm 0.001 \text{ t ha}^{-1}$) had the capacity to store lowest SOC (Table S1 and Fig. 6).

However, in cropland the SOC-S increased by 2.78% for AC and 13.51% for PC over time (from 2009 to 2015), increasing the SOC-S storage from $0.036 \pm 0.001 \text{ t ha}^{-1}$ to $0.037 \pm 0.001 \text{ t ha}^{-1}$ for AC, $0.037 \pm 0.001 \text{ t ha}^{-1}$ to $0.042 \pm 0.001 \text{ t ha}^{-1}$ for PC, respectively (Table S1). Also, this study reveals that the SOC-S storage insignificantly increased within cropland and over time but it's contradictory only for 2015 where PC and AC significantly ($p < 0.001$) different between them (Fig. 6).

3.6. Soil organic carbon stock within grassland types and over time

Overall, the SOC-S differed across the grassland types and over time which was significant ($p < 0.001$) within grassland types and between years, except spontaneously re-vegetated surfaces and shrubland without tree cover (Table S2 and Fig. 7). In 2009, the storage of SOC-S was insignificant among the five grassland types with the highest in the grassland without tree/shrub cover (GWOTC) ($0.053 \pm 0.003 \text{ t ha}^{-1}$), followed by grassland with sparse tree/shrub cover (GWSTC) ($0.051 \pm 0.005 \text{ t ha}^{-1}$), shrubland with sparse tree cover (SWSTC) ($0.049 \pm 0.008 \text{ t ha}^{-1}$), shrubland without tree cover (SWOTC) ($0.036 \pm 0.006 \text{ t ha}^{-1}$) and spontaneously re-vegetated surfaces (SRVS) ($0.035 \pm 0.002 \text{ t ha}^{-1}$), respectively. On the other hand, in 2015, SOC-S storage was significantly ($p < 0.001$) different within grassland types, except shrubland with sparse tree cover which was insignificant with other grassland types. Also, the highest and lowest amount of SOC-S was associated with grassland with sparse tree/shrub cover ($0.083 \pm 0.015 \text{ t ha}^{-1}$) and spontaneously re-vegetated surfaces ($0.035 \pm 0.015 \text{ t ha}^{-1}$), respectively (Table S2 and Fig. 7).

However, in grassland, the SOC-S increased by 62.75% for grassland with sparse tree/shrub cover, 30.19% for grassland without tree/shrub cover, 18.37% for shrubland with sparse tree cover, 11.11% for shrubland without tree cover and 8.57% for spontaneously re-vegetated surfaces over time (between 2009 and 2015), increasing the SOC-S storage

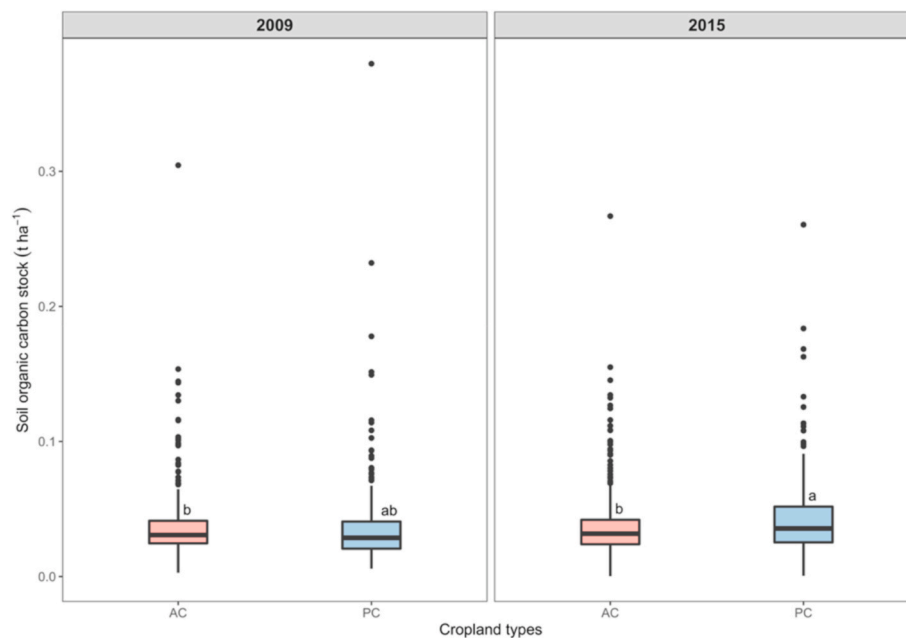


Fig. 6. The integrated box plot shows average SOC stocks (t ha^{-1}) in major cropland types [Annual crops (AC) and Perennial crops (PC)] in Italy. The black vertical line of the box plot indicates the median, while the bottom and top of the box are always the 25th and 75th percentiles. The ends of the box reflect the minimum and maximum of all the data. Also, different letters denote the significant differences at $p < 0.001$ within cropland types and years.

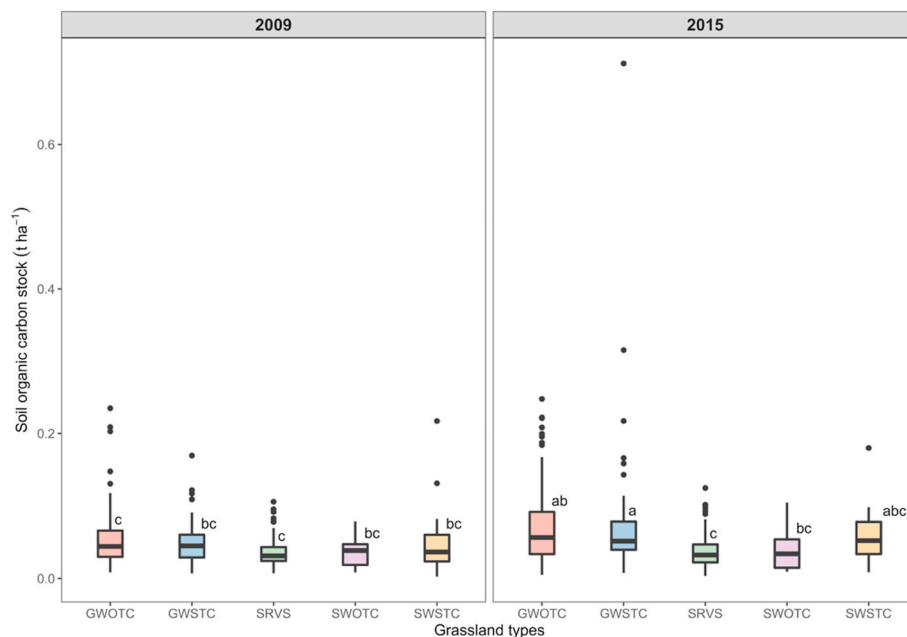


Fig. 7. The integrated box plot shows average SOC stocks (t ha^{-1}) in major grassland types [Grassland without tree/shrub cover (GWOTC), Grassland with sparse tree/shrub cover (GWSTC), Spontaneously re-vegetated surfaces (SRVS), Shrubland without tree cover (SWOTC) and Shrubland with sparse tree cover (SWSTC)] in Italy. The black vertical line of the box plot indicates the median, while the bottom and top of the box are always the 25th and 75th percentiles. The ends of the box reflect the minimum and maximum of all the data. Also, different letters denote the significant differences at $p < 0.001$ within crop-land types and years.

from $0.051 \pm 0.005 \text{ t ha}^{-1}$ to $0.083 \pm 0.015 \text{ t ha}^{-1}$ for grassland with sparse tree/shrub cover, $0.053 \pm 0.003 \text{ t ha}^{-1}$ to $0.069 \pm 0.003 \text{ t ha}^{-1}$ for grassland without tree/shrub cover, $0.049 \pm 0.008 \text{ t ha}^{-1}$ to $0.058 \pm 0.007 \text{ t ha}^{-1}$ for shrubland with sparse tree cover, $0.036 \pm 0.006 \text{ t ha}^{-1}$ to $0.040 \pm 0.007 \text{ t ha}^{-1}$ for shrubland without tree cover and $0.035 \pm 0.007 \text{ t ha}^{-1}$ to $0.038 \pm 0.004 \text{ t ha}^{-1}$ for spontaneously re-vegetated surfaces, respectively (Table S2). Also, this study shows that the SOC-S storage insignificantly increased within three grassland types (SRVS, SWOTC, and SWSTC) but significantly ($p < 0.001$) increased within others (GWSTC and GWOTC) grassland, over time (Fig. 7).

3.7. Soil organic carbon stock within forest land types and over time

Comparing the different forest land types, overall, the largest SOC-S was encountered in the mixed woodland compared with broadleaved

woodland and coniferous woodland which was significantly ($p < 0.001$) different within forest land types but insignificant over time (Table S3 and Fig. 8). In 2009, the highest storage of SOC-S was recorded in the coniferous woodland (CW) ($0.089 \pm 0.021 \text{ t ha}^{-1}$), followed by mixed woodland (MW) ($0.064 \pm 0.008 \text{ t ha}^{-1}$), and the lowest was recorded in the broadleaved woodland (BW) ($0.056 \pm 0.002 \text{ t ha}^{-1}$) forest which was insignificant. In contrast, in 2015, a significant ($p < 0.001$) difference in SOC-S storage was observed between broadleaved woodland ($0.064 \pm 0.002 \text{ t ha}^{-1}$) and mixed woodland ($0.084 \pm 0.007 \text{ t ha}^{-1}$), but coniferous woodland was insignificant with other forest land types (Table S3 and Fig. 8).

However, in forest land the SOC-S increased by 14.29% for broadleaved woodland and 31.25% for mixed woodland over time (from 2009 to 2015), increasing the SOC-S storage from $0.056 \pm 0.003 \text{ t ha}^{-1}$ to $0.064 \pm 0.002 \text{ t ha}^{-1}$ for broadleaved woodland, $0.064 \pm 0.008 \text{ t ha}^{-1}$ to

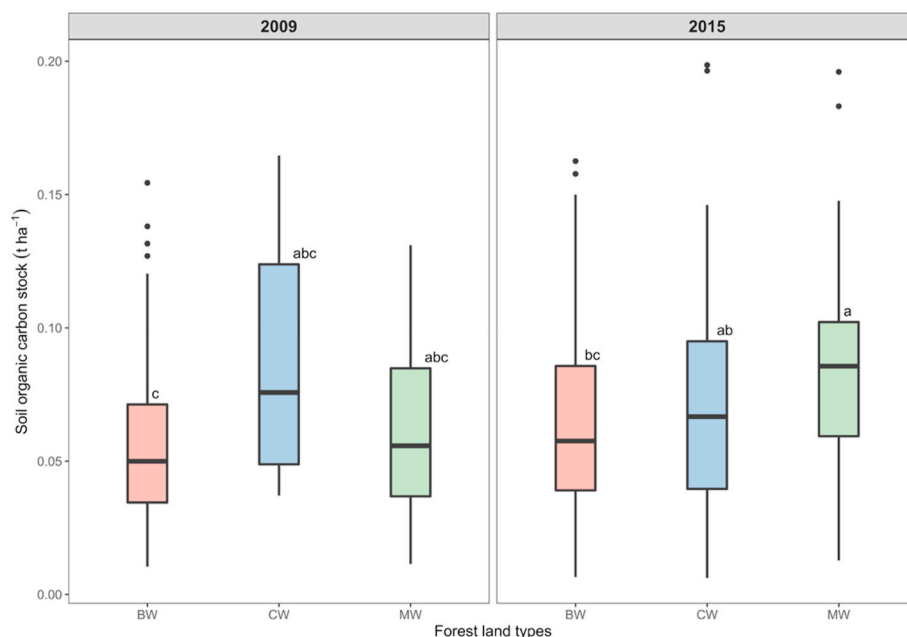


Fig. 8. The integrated box plot shows average SOC stocks (t ha^{-1}) in major forest land types [Broadleaved woodland (BW), Coniferous woodland (CW), and Mixed woodland (MW)] in Italy. The black vertical line of the box plot indicates the median, while the bottom and top of the box are always the 25th and 75th percentiles. The ends of the box reflect the minimum and maximum of all the data. Also, different letters denote the significant differences at $p < 0.001$ within cropland types and years.

$0.084 \pm 0.007 \text{ t ha}^{-1}$ for mixed woodland, respectively. But the SOC-S storage decreased by 20.22% for coniferous woodland from $0.089 \pm 0.021 \text{ t ha}^{-1}$ to $0.071 \pm 0.004 \text{ t ha}^{-1}$ between 2009 and 2015 (Table S3). Also, this study reveals that the SOC-S storage insignificantly increased and decreased within forest land and over time (Fig. 8).

3.8. Soil organic carbon stock in different land use types and over time

Among the three land use types (LUT), the storage of SOC-S varied across the LUT and over time which was significantly ($p < 0.001$) different within LUT but not for over time except grassland (Table S4 and Fig. 9a). In 2009, SOC-S was significantly ($p < 0.001$) higher in forest land ($0.059 \pm 0.003 \text{ t ha}^{-1}$), compared with grassland ($0.047 \pm 0.002 \text{ t ha}^{-1}$) and cropland ($0.036 \pm 0.001 \text{ t ha}^{-1}$). By contrast, in 2015, a significant ($p < 0.001$) difference in SOC-S storage was observed for all land use, except for forest land and grassland which was insignificant. Also, it is true that forest land had the capacity to store the largest ($0.067 \pm 0.002 \text{ t ha}^{-1}$) soil organic carbon among the different land use types, and cropland showed the lowest ($0.036 \pm 0.001 \text{ t ha}^{-1}$) soil carbon storage capacity (Table S4 and Fig. 9a).

However, the SOC-S storage increased by 8.33% for cropland, 13.56% for forest land, and 29.79% for grassland between 2009 and 2015, increasing the SOC-S storage from $0.036 \pm 0.001 \text{ t ha}^{-1}$ to $0.039 \pm 0.001 \text{ t ha}^{-1}$ for cropland, $0.059 \pm 0.003 \text{ t ha}^{-1}$ to $0.067 \pm 0.002 \text{ t ha}^{-1}$ for forest land, $0.047 \pm 0.002 \text{ t ha}^{-1}$ to $0.061 \pm 0.002 \text{ t ha}^{-1}$ for grassland, respectively (Table S4). Also, this study indicates that the SOC-S storage significantly ($p < 0.001$) increased in grassland over time but not for cropland and forest land which also follow the increasing trend but insignificantly (Fig. 9a).

Over time (six-year period), the amounts of SOC-S in the different LUT increased in the following order: grassland > forest land > cropland, with total SOC-S storage in each land use being 0.014 t ha^{-1} , 0.008 t ha^{-1} , and 0.003 t ha^{-1} , respectively. But the mean SOC-S storage of the different LUT was in the following order: forest land > grassland > cropland for both years 2009 and 2015 (Table S4 and

(b)

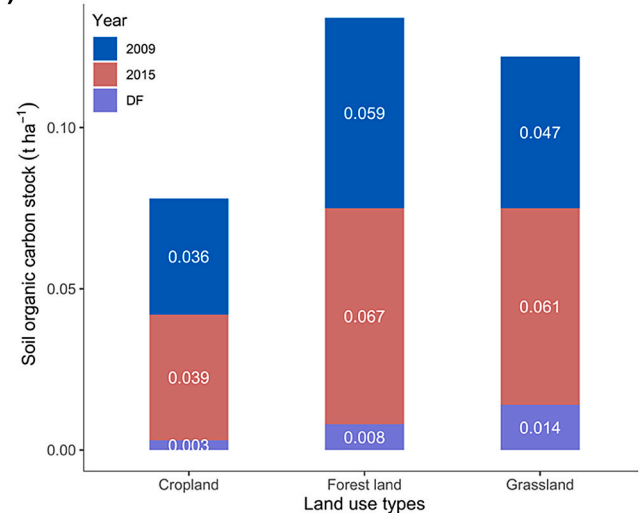


Fig. 9b. The integrated stack bar plot shows the SOC stocks (t ha^{-1}) in major land use types over time (2009 and 2015), and the differences (DF) between 2009 and 2015.

Fig. 9b).

3.9. Relationship of SOC stock with soil properties and different years

SOC stock (SOC-S) was positively correlated with SOC content, coarse particle, sand and silt percentage, rock fragments, phosphorus, nitrogen and elevation, but negatively correlated with bulk density, soil pH, free CaCO_3 and clay percentage ($p \leq 0.05$) (Fig. 10). Also, different years indicate a significant positive correlation with SOC stock (SOC-S) where SOC strongly correlated with SOC-S. The measured soil variables

(a)

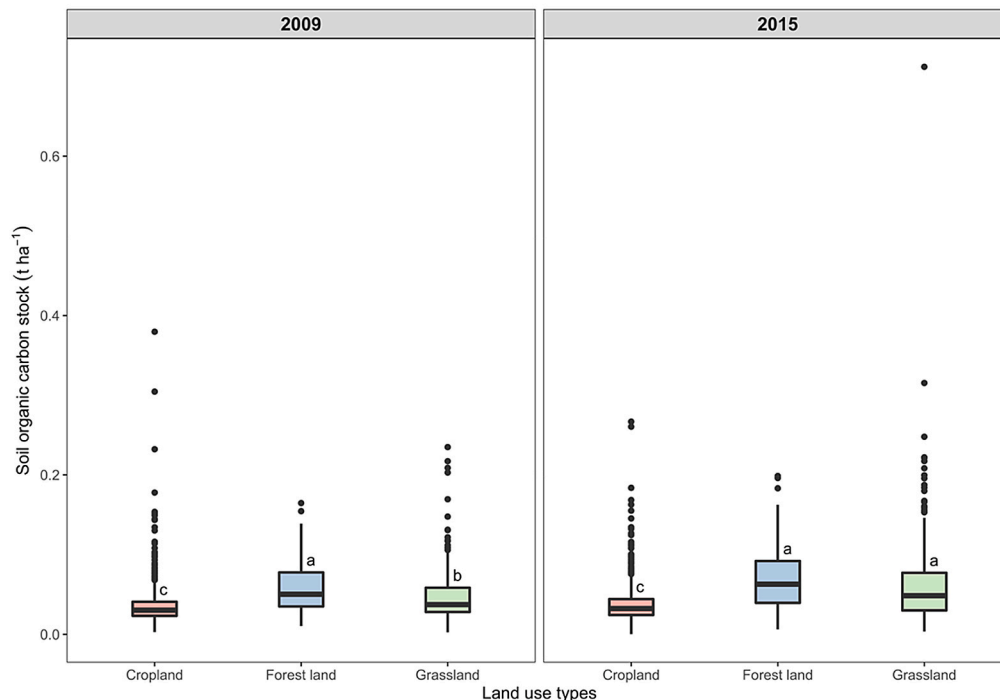


Fig. 9a. The integrated box plot shows average SOC stocks (t ha^{-1}) in major land use types in Italy. The black vertical line of the box plot indicates the median, while the bottom and top of the box are always the 25th and 75th percentiles. The ends of the box reflect the minimum and maximum of all the data. Also, different letters denote the significant differences at $p < 0.001$ within cropland types and years.

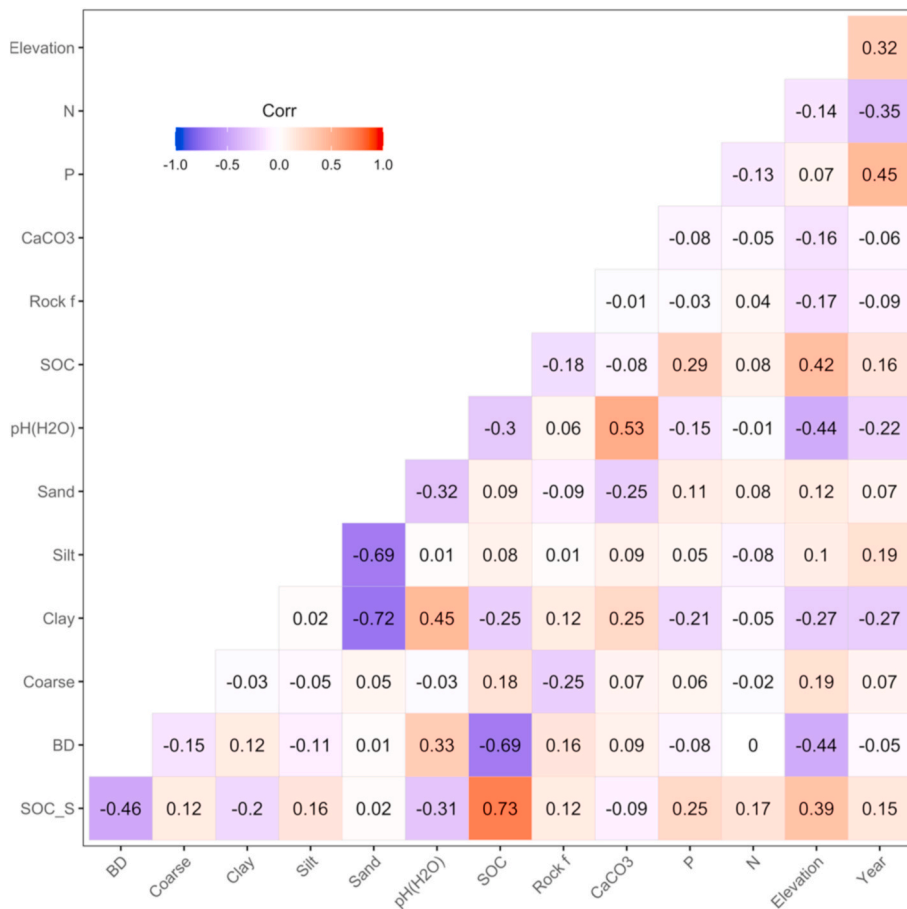


Fig. 10. Correlation matrix among SOC-S and other physical and chemical properties of soil. The number of each block reflects the Spearman's rank correlation coefficients at $p \leq 0.05$, with red and violet colour representing positive and negative correlations, respectively. The correlation coefficient is statistically insignificant, as indicated by the white block. The soil properties include: pH(H₂O) = Soil pH, Coarse = Coarse particle, BD = Bulk density, CaCO₃ = Free CaCO₃ percentage, Sand = Sand percentage, Silt = Silt percentage, Clay = Clay percentage, SOC = Soil organic carbon, Rock f = Rock fragments (%), P = Soil phosphorus, N = Soil nitrogen, SOC_S = Soil organic carbon stock, Elevation = Elevation and Year = Years. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

also displayed a significant positive and negative relationship among themselves.

Principal component analysis (PCA) has been used with fourteen variables to build and extract the smallest number of subsets feasible to explain the variation in the dataset (Fig. 11). The PCA result indicates

that the first three principal components (PCs) explained the majority of the variance in the dataset (54.30 percent), with the first two PCs accounting for 44.20 percent of the total variation with an eigenvalue larger than 1. The most relevant component (PC1) explained 27.80 percent of the variance in SOC and elevation with the maximum loading

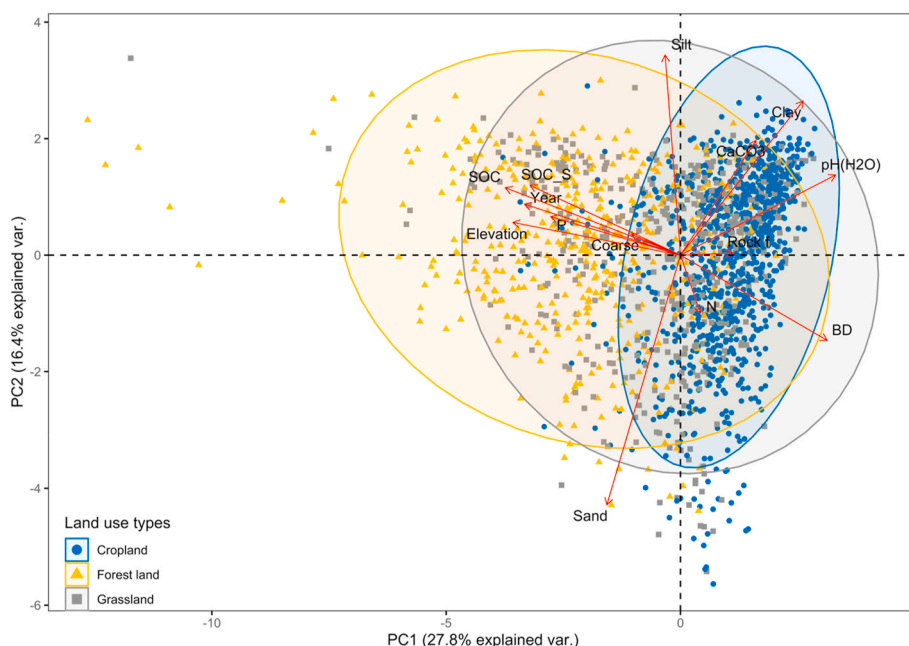


Fig. 11. Principle component analysis (PCA) biplot of soil physical and chemical parameters as vectors ($n = 14$) with Italian regions coloured coded as three land use types ($n = 1891$). The soil physical properties included: BD = Bulk density, Coarse = Coarse particle, Sand = Sand percentage, Silt = Silt percentage, Clay = Clay percentage, pH(H₂O) = Soil pH, Rock f = Rock fragments (%), and the chemical soil properties comprised: P = Soil phosphorus, N = Soil nitrogen, SOC = Soil organic carbon, SOC_S = Soil organic carbon stock, Elevation = Elevation and CaCO₃ = Free CaCO₃ percentage; and Year. Here, perpendicular direction signifies an uncorrelated relationship, but the opposite vectors and small-angle vectors represent negative and positive correlated vectors, respectively.

larger than >0.35 . The second component (PC2), on the other hand, revealed larger loadings for sand percentage, silt percentage and nitrogen content with 16.40 percent of variance explained (Table S5). As SOC and elevation are highly correlated with each other ($r = 0.42$, $p \leq 0.05$) (Fig. 10), the variable with the largest loading, SOC was considered from the first component for the SOC–S estimation in different LUT in Italy. Similarly, silt percentage was discarded with N and therefore, nitrogen and sand percentage were selected from the PC2.

4. Discussion

4.1. Prediction performance of pedotransfer functions

In theory, soil BD is easy to be measured and does not need complex analysis. But in practice, large-scale direct measurement of soil BD remains labour-intensive, time-consuming, tedious and costly (Kaur et al., 2002). As a result, BD is usually absent from normal soil surveys which was also true for the LUCAS database. Thus, the missing BD must be obtained by alternative methods called pedotransfer functions (PTFs) (Martin et al., 2009; Morvan et al., 2008; Suuster et al., 2011). Also, it should be noted that in this study using PTFs, is based only on SOC for forest land while SOC and texture for grassland and cropland, respectively (Adams, 1973; Chiti et al., 2018). In this study, using PTFs for calculating BD, it can be observed that only cropland and grassland, tend to overestimate, considering the mean of MDs (Fig. 2). These findings are aligned with several previous research (Han et al., 2012; Nasta et al., 2020; Xiangsheng et al., 2016). But forestland shows a tendency to underestimate. Note, however, that the increase of the error could also be due to a higher number of samples in forest land in comparison to cropland and grassland (De Vos et al., 2005; Lilly and Lin, 2004).

4.2. Soil organic carbon, bulk density and land use

Soil bulk density (BD) is a critical soil physical property that is influenced by a variety of circumstances. Several studies have revealed that BD has been linked to SOC and texture (Federer, 1983; Kaur et al., 2002), and it changes with loss-on-ignition (LOI) (De Vos et al., 2005), depth (Huntington et al., 1989), pH, and cation exchange capacity (CEC) (Benites et al., 2007; Han et al., 2012; Tranter et al., 2007). We found that BD was strongly related to SOC and was influenced by the texture (sand, silt, and clay) and pH of the soils in the study region (Figs. 10 and 11). SOC has a dominant effect on BD in uncultivated soils, such as forest soils, due to its lower specific gravity than mineral soil particles and its aggregation effects on soil structure (Adams, 1973; Xiangsheng et al., 2016).

SOC and BD relationships are commonly used to predict SOC stock (Post et al., 1982; Post and Kwon, 2000). The importance of such relationships is demonstrated in this study by the fact that BD tends to increase as SOC concentration decreases, which is significant ($p < 0.001$) for all land uses (more significant for forest land than grassland and cropland) and for different years (Fig. 3). Several authors found that land management practices affect BD (Céspedes-Payret et al., 2016; Munny et al., 2021), which agreed with our findings. However, the prevalent assumption is that decreasing SOC content reduces aggregate stability and consequently soil porosity, accompanied by a rise in BD values. Conversely, using similar linear logic, it is conceivable to foresee an opposite process: beginning with SOC gain, aggregate stability, and as a result, soil porosity rise, resulting in a fall in BD values (Ali et al., 2017; Morisada et al., 2004). There is a perception that cultivation degrades soil organic matter and decreases the inherent stability of soil aggregates, making them vulnerable to water and wind damage and resulting in compacted soil layers with increasing bulk densities. Other research has found that in some soils, packing arrangement is more significant than SOC quantity (Loveland and Webb, 2003), and SOC distribution is more relevant than SOC quantity. (Céspedes-Payret et al., 2016). But results of other studies showed no link between SOC levels

and aggregate stability (Carter et al., 1994).

4.3. Response of SOC dynamics with environmental variables

Among the most important topics in climate change research concerns the future dynamics of a substantial amount of C that is now stored in terrestrial ecosystem as SOC. Will the quantity of C in this pool rise or fall as the planet warms? The SOC's role as a source or sink is mostly determined by regional temperature and precipitation (Poll et al., 2013; Zhang et al., 2019). According to our findings, SOC storage rose with rising mean annual precipitation (MAP) and decreased with increasing mean annual temperature (MAT) (Fig. 4). Because of the decline in SOC storage, a large amount of carbon would be released from the soil into the atmosphere, expediting the process of global warming. But land use and management have a greater influence on SOC dynamics than climate factors (Fantappiè et al., 2011; Wang et al., 2018), which is consistent with our findings (Fig. 4).

However, SOC dynamics is a complicated biogeochemical process driven by a range of direct and indirect variables like as precipitation, temperature, and carbon inputs (Jia et al., 2015; Sierra et al., 2015). To being with, higher temperatures increase microbial activity, causing more SOC to be degraded by microorganisms and therefore depleting SOC storage. Second, raising the temperature stimulates plant growth, which results in more SOC storage. As a result, the lower SOC storage with MAT might be explained by a larger temperature sensitivity of microbial breakdown compared to plant C inputs. In addition, the dynamics of SOC storage were affected by soil properties which is also true for this study (Figs. 10 and 11). Plant growth and microbial activity would be influenced by changes in soil properties, resulting in the regulation of soil properties on SOC storage (Tan et al., 2020; Tang et al., 2018). In this study, the SOC storage following a similar trend for all LUT, which is consistent with earlier studies (Jia et al., 2015; Wang et al., 2013). According to some research, a 1 °C increase in air temperature can result in an increase in C emission of 10–28%, or 11 to 34 Pg C year⁻¹ (Schimel et al., 1994). Furthermore, Gabarrón-Galeote et al. (2015) found that the most important factors influencing organic C inputs into the soil is precipitation. Changes in precipitation patterns have a great potential to influence SOC storage and dynamics (Gerten et al., 2008), because water is an essential driver of nearly all chemical and biological processes, including plant growth and survival, photosynthesis, microbial activity, and soil respiration, all of which are important controlling factors in SOC cycling (Aanderud et al., 2010; Chen et al., 2016). A meta-analysis of the precipitation manipulation studies revealed that increased precipitation enhances soil respiration by 30% on average, while decreased precipitation inhibits soil respiration by 12% (Wu et al., 2011).

Environmental factors that can influence SOC differ depending on the scale, according to a number of studies (Hobley et al., 2015; Longbottom et al., 2014; Zhao et al., 2017). On a global and regional scales, temperature and precipitation are the most important factors influencing SOC storage, and SOC rises as precipitation increases and temperature decreases (Follett et al., 2012; Herold et al., 2014). Another research discovered that rising temperature and precipitation had an impact on soil respiration, which is a significant factor in determining how much SOC is stored (Chen et al., 2013). Including, a study of Mediterranean woodlands found that increased summer soil moisture accelerates C cycling by accelerating annual stem primary production, litterfall, soil respiration, and net annual plant-derived C input to the soil (Cotrufo et al., 2011), all of which are connected to SOC dynamics.

We found a significant linear correlation between SOC–S storage and elevation, as well as different land uses, in this study, showing that higher altitudes have more SOC–S storage capacity. Also, our results indicate that there is generally SOC–S storage increased significantly in forest land and grassland but not for cropland (Fig. 5). Several previous studies have been suggested that the larger SOC–S storage at higher elevation due to the slower decomposition rates of SOC (Dieleman et al.,

2013; Zinn et al., 2018). Climate change, particularly a drop in temperature, has been blamed in many studies for the decrease (Salinas et al., 2011; Sousa Neto et al., 2011). Other research has suggested that adverse soil conditions (e.g., waterlogged soil, low pH values, high toxicity of Al^{3+} and Fe^{2+} concentrations) may be the main driver for slower SOC breakdown at higher elevations, restricting soil fauna and microbial activity (Dieleman et al., 2013).

4.4. Overview of SOC stock change over time as affected by land use and soil management

The SOC content can either be increased or decreased depending on various factors such as soil type, climate variable, topography, functional diversity and microbial activity of soils, and soil management practices (Post and Kwon, 2000; Sheikh et al., 2009). Also, SOC is greatly influenced by vegetation through OM input, and therefore LUC is one of the most important factors which influences SOC stock build-up. For example, it was reported that the conversion of forest land to cropland led to a decrease in the quality of soil owing to the reduced SOC stocks (Guo and Gifford, 2002; Murty et al., 2002). However, our research did not take into consideration of many other factors which are responsible for the gain or loss of SOC stock such as soil types, microbiological and biochemical properties of soil, etc. In this study, results showed that LUC and soil management practices can significantly ($p < 0.001$) influences SOC dynamics and SOC storage from the soil and varied among LUC types but not for over time except grassland (Fig. 9a). Spatially, the mean SOC-S storage of the different LUC's were in the following order: forest land > grassland > cropland for both years 2009 and 2015 (Table S4). This finding is consistent with other literatures that show LUC's and management practices have a direct impact on soil physical and chemical properties, as well as significant impacts on SOC stock (Ali et al., 2017; Anokye et al., 2021; Deng et al., 2014; Eid et al., 2017). On the other hand, the SOC-S storage increased by 8.33% for cropland, 13.56% for forest land, and 29.79% for grassland during the period of 2009–2015, while SOC-S storage increased significantly ($p < 0.001$) in grassland over time but not for cropland and forest land which also follow the increasing trend but insignificantly (Fig. 9a), as several authors have suggested (Ali et al., 2017; Wang et al., 2018).

The fact that forest land has a higher SOC-S storage capacity than grassland and cropland might be due to the lower OM inputs in the latter systems. Cropland's, annual and perennial agriculture disturbs soils more than grassland and forest land, explaining its lower SOC-S (Anokye et al., 2021; Logah et al., 2020). Less soil disturbance tends to conserve more SOC-S, and mixed species forest are more robust and better systems for SOC-S storage, according to Morris et al. (2007). Also, higher amount of SOC-S storage in forest land may be because of higher leaf litter and extensive root systems of forest trees as well as zero perturbation and modification of hydro-thermal regime in these soils, what is the primary source of SOC (Ali et al., 2017; Badalamenti et al., 2017; Rees et al., 2005). Litter decomposition in a forest ecosystem is a complex process caused by the combination of abiotic and biotic variables such as temperature, moisture, and community features (Liu et al., 2016; Takahashi et al., 2010). Complex molecules are predicted to be preferentially stabilized as SOC based on the widespread assumption that the chemical composition of the litter has a substantial effect in its rate of degradation. As a result, a direct relationship between litter dynamics and soil C quality has been proposed and recognized in SOC-S storage, which might explain why forest land has increased SOC stock. Fibrous root system of the grassland soils and no-tillage condition also increased the SOC-S storage of the soils (Conant et al., 2017; Wang et al., 2018) which is true for this study. There are large variations in SOC-S storage among grassland types, although grassland tends to result in SOC-S accumulation over time, which was also significant. Considerable rates of SOC-S storage in Italian grassland increase (0.014 t ha^{-1}) significantly in contrast to forest land (0.008 t ha^{-1}) and cropland (0.003 t ha^{-1}), respectively during the period of 2009–2015, suggesting

that grasslands also have the potential capacity to sequester large amounts of atmospheric carbon in the soil, which is consistent with previous studies (Conant et al., 2017; Nakagami et al., 2009; Yang et al., 2010). In contrast, the decreased SOC-S in cropland may be attributed to frequent cultivation, crop residue clearance, and crop residue burning during land preparation (Ashagrie et al., 2007; Toru and Kibret, 2019). Moreover, intensive management and cultural practices in cropland accelerated the turnover rates of macro aggregates and lead to destabilization of the labile soil organic C compounds (Ashagrie et al., 2007; Lipper et al., 2011). Additionally, if cereal cover crops have been used, the burial of fresh biomass C with ploughing accelerated SOC stock depleting, most presumably because of priming effects, revealing that C inputs management is imperative for its accumulation in agricultural production systems. Other researchers also reported that tillage, alters soil moisture, aeration, and nutrient concentration, resulting in increased oxidation of soil organic matter and loss via soil microbes, which can restrict organic matter inputs from above and below-ground plants (Yimer et al., 2006; Zhao et al., 2005). Soil may therefore be a source or sink of atmospheric carbon depending on LUC's and soil management practices.

4.5. Soil physical and chemical properties in relation with SOC stocks

In terms of soil properties, our result showed that SOC-S storage is positively correlated with the SOC content, nitrogen, phosphorus, soil texture, and elevation (Fu et al., 2010; Li et al., 2013; Liu et al., 2016), but negatively correlated with BD, soil pH, free CaCO_3 , which is consistent with other findings (McGrath and Zhang, 2003; Omar et al., 2017; Wang et al., 2018). In this study, SOC-S storage was shown to be inversely related to clay content at the surface soil layer (Figs. 10 and 11) which is consistent with those reported by Jobba'gy et al. (2000). Some research has found that C stability mechanisms may differ between topsoil and deep soil, as well as their C dynamic processes and response to external environmental change (Liu et al., 2016, 2019; Zhang et al., 2019) which is compatible with our findings. The SOC-S storage in the surface soil were mainly affected by precipitation and temperature with elevation, while clay percentage was the best predictor in deeper layers (Jobba'gy et al., 2000).

5. Conclusion

SOC provides essential ecosystem functions such as nutrient cycling, C sequestration, and climate regulation in the terrestrial biosphere, which can mitigate or accelerate climate change. Also, SOC dynamics represent a persistent source of uncertainty in our understanding of the global carbon cycle. So, monitoring carbon stocks and their dynamics over time is an important pre-requisite for reducing the uncertainty of carbon balance measurements and identifying appropriate management practices and time scales for changing the carbon loss trend in the country level, using a reliable soil database like LUCAS. This study showed that SOC is greatly influenced by vegetation through OM input, and therefore one of the most significant factors controlling SOC stock build-up over time is LUC. Although vegetation types and their biochemical structure influence the SOC stocks and dynamics in surface soils, but it is true that climate variables (temperature and precipitation) and elevation also significantly contribute the SOC storage and dynamics in superficial layer of soil over time in Italy. In terms of soil properties, our result highlighted that, SOC stocks are directly correlated with phosphorus and nitrogen, but inversely correlated with BD, soil pH and free CaCO_3 . However, our research did not take into consideration many other factors which are responsible for the gain or loss of SOC stock such as soil types, microbiological and biochemical properties of soil, etc. Further research is needed to clarify the precise cause of this.

In conclusion, we can say that LUC and soil management practices coupled with climate variables can significantly influences SOC storage and its dynamics in Italy. And, SOC storage increased by 8.33% (0.036 t

ha⁻¹ to 0.039 t ha⁻¹) for cropland, 13.56% (0.059 t ha⁻¹ to 0.067 t ha⁻¹) for forest land, and 29.79% (0.047 t ha⁻¹ to 0.061 t ha⁻¹) for grassland during the years of 2009–2015, which indicates the importance of monitoring SOC stocks for mitigating climate change. Thus, this country-level baseline information may be utilized as a valuable tool to monitor the future trends in SOC storage in the context of projected climate conditions coupled with LUC scenarios. The study also demonstrates the importance of a carbon-friendly land use strategy and sustainable soil management practices to enhancing the SOC storage capacity of different land use systems to mitigate greenhouse gas emissions and combat climate change.

Credit author statement

Md. Zulfikar Khan: Conceptualization, Data curation, Formal data analysis, Visualization, Writing – original draft & editing. **Tommaso Chiti:** Investigation, Validation, Writing – review & editing, Supervision.

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Data availability

The datasets and R-code used to support the findings of this study are available from the corresponding author on reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2022.114452>.

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